AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2011

Biology Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer ALL NINE questions. Questions 7, 8 and 9 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 4 Marks)

a. State the role of the following in the welfare of mankind. (2 Marks)

i. Horticulturist

_____

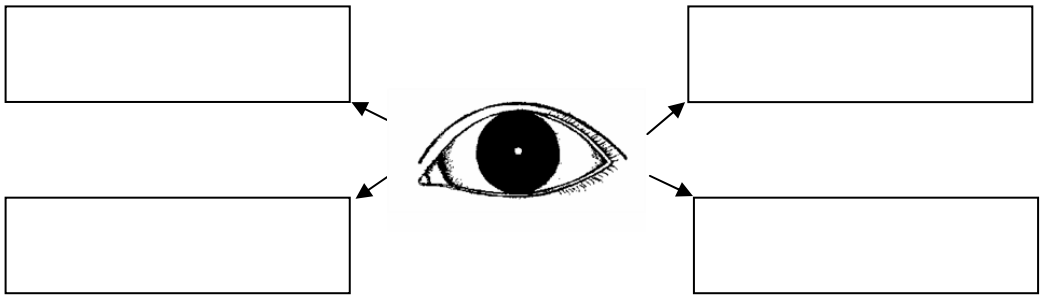
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ii. Surgeon

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_____

b. Complete the given diagram by naming the different tissues that make up the eye as an organ. (2 Marks)



Q.2. (Total 4 Marks)

a. How is the validity of a hypothesis tested? (1 Mark)

b. The scientific name of the frog is rana tigrina. Why do we give a scientific name to a living organism? Rewrite the given scientific name of the frog following the rules of binomial nomenclature. (3 Marks)

Q.3. (Total 4 Marks)

a. The Golgi bodies in an animal cell have lost their ability to produce lysosomes. What changes do you expect in such a cell? (2 Marks)

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b. Define the term haploid. Give an example of haploid cells found in the human body. (2 Marks)

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Q.4. (Total 4 Marks)

a. A very small amount of papaya is often added to meat to tenderise it (make it soft) as papaya contains some enzymes.

Write TWO characteristics of enzymes that are referred in the above example. (2 Marks)

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b. The table shows the difference in the amount of energy produced as a result of aerobic and anaerobic respiration. (2 Marks)

	Aerobic respiration	**Anaerobic respiration**
Amount of energy	36 ATP	2 ATP

Give a reason for the difference in values in the above table.

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Q.5. (Total 4 Marks)

a. Name any TWO micro-nutrients required by a plant for its normal growth and reproduction. (1 Mark)

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b. The table shows the composition of diet of different vulnerable groups. (3 Marks)

Groups	Diet
Child (4-6 years)	Meat, eggs, bread and potato
Adult (above 50)	Fish, apples, rice and spinach
Labourer	

i. Why are the dietary requirements of a child different from those of an adult?

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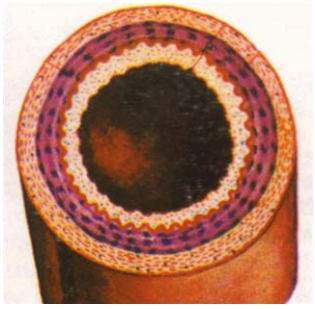
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ii. Complete the given table by inserting the dietary requirement of a labourer.

Q.6. (Total 5 Marks)

a. Transpiration causes loss of water, yet this loss has some benefits. Describe TWO ways in which transpiration is beneficial for the plants. (2 Marks)

b. The diagram shows a type of blood vessel. (3 Marks)

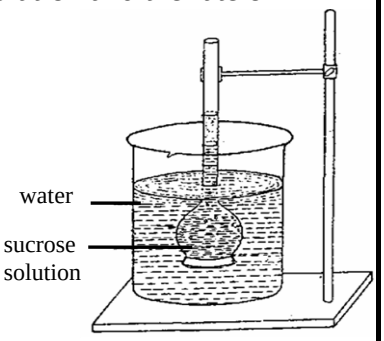


Identify the blood vessel shown in the diagram. Give TWO characteristics of this blood vessel that are visible in the diagram.

Q.7. (Total 5 Marks)

EITHER

a. The diagram shows the set up to demonstrate the process of osmosis using sucrose solution. The experiment was performed using different concentrations of sucrose solution and the rate of osmosis was observed.



Present the results given in the table in the form of a graph.

(Consider the position of quantities on X-axis and Y-axis)

S.No.	Concentration of Sucrose Solution (%)	Rate of osmosis (arbitrary units)
1	2	10
2	3	20
3	5	40
4	6	50
5	8	70

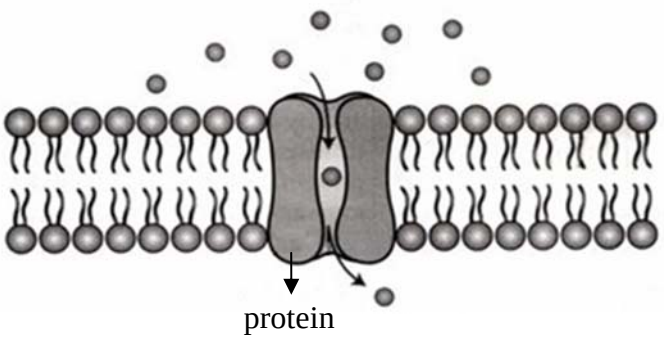
OR

b. Biodiversity in Pakistan is facing a great loss due to many human activities. Give your opinion with reference to some human interventions that result in life threatening effects on animals.

Q.8. (Total 5 Marks)

EITHER

a. The process shown below is a type of passive transport.



Justify this process as passive while describing the importance and mechanism of the transport system.

OR

b. Explain the role of mitosis and meiosis in the continuity of living organisms.

Q.9. (Total 5 Marks)

EITHER

a. With the help of graph (not to scale) explain the effect of the following factors on the activity of enzymes:

i. Temperature
ii. Concentration of substrate

OR

b. Why is ATP regarded as the currency of a cell? Explain the formation of an ATP molecule from adenosine diphosphate.

END OF PAPER